

STEPHENS COUNTY AP, TX, USA

WMO: 720578

Lat: 32.717N

Lon: 98.892W

Elev: 1283

StdP: 14.03

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00176

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.9	25.6	5.7	7.7	31.3	10.2	9.7	35.9	25.4	57.0	23.7	56.8	6.1	350	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.2	103.7	71.1	100.9	71.4	99.2	71.3	76.5	89.9	75.5	89.0	74.8	88.6	9.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.3	130.0	79.2	72.5	126.2	79.5	71.6	122.3	79.8	40.8	89.6	39.9	88.2	39.0	88.4	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.8	20.6	18.8	DB	13.9	106.0	3.8	2.3	11.2	107.6	9.0	109.0	6.9	110.3	4.1	111.9	
			WB	13.0	78.3	4.3	1.3	9.9	79.2	7.4	80.0	5.0	80.7	1.9	81.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.6	46.9	51.4	60.5	68.1	74.7	83.4	86.4	86.6	79.0	68.1	56.9	48.4
	DBStd	16.42	10.18	12.34	10.23	7.96	7.60	4.72	4.40	4.97	6.61	8.98	9.66	10.26
	HDD50	538	181	122	29	0	0	0	0	0	0	6	45	155
	HDD65	2116	564	401	209	59	17	0	0	0	1	74	271	520
	CDD50	6965	84	161	353	543	766	1003	1127	1134	869	566	252	107
	CDD65	3069	3	20	68	152	318	553	662	669	420	169	29	6
	CDH74	38872	75	220	689	1650	3532	7180	9220	9338	4841	1794	260	73
	CDH80	20842	11	60	210	640	1637	3959	5457	5654	2469	674	59	12
(14) Wind	WSAvg	8.9	8.6	9.4	10.3	11.0	10.1	9.6	8.3	7.8	7.1	8.4	8.7	8.2
(15) Precipitation	PrecAvg	31.4	1.6	2.0	2.6	2.8	3.9	3.5	2.1	2.5	3.0	2.9	2.4	1.6
	PrecMax	45.3	4.5	7.8	7.1	9.3	8.3	12.2	5.2	7.4	6.6	6.8	7.8	8.3
	PrecMin	15.7	0.1	0.0	0.0	0.2	1.9	0.6	0.0	0.0	0.4	0.2	0.2	0.0
	PrecStd	7.7	1.1	1.8	1.7	2.0	2.0	2.6	1.5	1.7	1.7	1.9	2.0	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.6	87.5	90.2	97.2	100.5	104.4	106.2	106.2	102.2	94.3	86.3	81.3
		MCWB	56.3	61.2	62.3	63.6	71.6	71.1	70.3	70.4	70.5	67.2	64.3	59.6
	2%	DB	75.0	79.9	85.8	90.8	96.4	100.4	103.1	103.6	97.9	90.6	80.8	74.9
		MCWB	52.9	59.5	62.1	63.7	70.1	71.2	70.9	71.3	70.4	67.9	62.0	57.0
	5%	DB	69.6	75.2	81.5	87.1	91.8	97.8	100.2	101.1	95.1	86.9	76.7	70.2
		MCWB	51.9	58.4	60.7	64.2	68.7	71.5	71.5	71.5	70.1	66.5	60.5	55.6
	10%	DB	64.1	70.4	77.3	83.0	88.9	95.1	98.4	99.2	92.4	83.7	72.6	65.4
		MCWB	49.4	56.2	59.6	63.3	68.3	72.1	71.8	71.6	70.0	64.4	59.2	53.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.5	65.9	67.5	71.3	76.0	78.1	77.8	77.2	76.3	74.5	69.4	65.4
		MCDB	72.0	73.8	80.1	82.8	90.5	92.7	90.8	91.7	86.8	81.6	78.8	73.0
	2%	WB	56.6	64.0	66.0	69.7	74.0	76.1	76.4	75.5	74.8	71.7	66.6	61.9
		MCDB	67.4	73.1	78.3	81.6	87.6	89.8	89.7	90.9	84.9	83.2	74.3	68.7
	5%	WB	53.6	61.8	64.4	68.3	72.8	75.1	75.4	74.7	73.7	70.2	64.4	58.7
		MCDB	67.7	71.0	75.8	79.7	85.7	88.9	89.1	90.5	84.4	81.3	71.3	66.1
	10%	WB	50.9	58.9	62.9	66.9	71.6	74.1	74.6	73.7	72.9	68.4	62.2	54.9
		MCDB	62.8	68.9	73.3	77.8	83.6	88.1	88.6	90.1	84.2	79.2	70.2	64.4
(35) Mean Daily Temperature Range	5% DB	MDBR	24.4	23.1	24.2	24.0	22.3	21.1	21.4	22.2	21.6	24.3	23.4	21.9
		MCDBR	31.6	29.1	28.1	28.3	27.1	24.3	24.1	24.0	24.6	26.4	26.4	29.0
		MCWBR	16.2	13.9	12.1	12.1	8.7	6.4	6.0	5.8	6.3	10.6	12.7	15.4
	5% WB	MCDBR	29.5	24.8	22.0	21.7	21.2	20.4	19.7	21.5	19.2	21.4	20.7	22.0
		MCWBR	16.9	14.4	10.9	11.0	8.3	6.8	6.2	5.9	5.5	10.5	14.3	14.6
(40) Clear-Sky Solar Irradiance	taub		0.296	0.305	0.333	0.370	0.407	0.415	0.421	0.413	0.383	0.349	0.320	0.300
	taud		2.534	2.519	2.453	2.371	2.335	2.344	2.382	2.378	2.410	2.491	2.501	2.531
	Ebn at Noon		295	303	300	292	280	276	274	275	280	283	283	287
	Edh at Noon		26	30	34	39	41	40	38	38	35	30	27	25
(44) All-Sky Solar Radiation	RadAvg		982	1177	1551	1878	1987	2203	2194	1992	1673	1326	1036	861
	RadStd		104	173	108	150	198	183	139	135	133	199	131	104

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page